

ENGINEERING MATHEMATICS – M1 & M2 QUICK NOTES

Prepared for Mechanical Students – Simple English Revision Notes

Topics cover basics of Engineering Mathematics 1 and 2.

1. Matrices and Determinants (M1)

Matrix: Rectangular array of numbers.

Types: Row, Column, Square, Identity, Zero.

Determinant of 2x2:

$\begin{vmatrix} a & b \end{vmatrix}$

$\begin{vmatrix} c & d \end{vmatrix} = ad - bc$

Inverse of Matrix:

$A^{-1} = \text{adj}(A) / |A|$

Applications: Solving linear equations.

2. Differential Calculus

Derivative: Rate of change.

$$d(x^n)/dx = nx^{n-1}$$

$$d(\sin x)/dx = \cos x$$

$$d(\cos x)/dx = -\sin x$$

$$d(e^x)/dx = e^x$$

Applications:

- Tangent & Normal
- Maxima & Minima
- Velocity & Acceleration

3. Integral Calculus

Integration = Reverse of differentiation.

$$\int x^n dx = x^{n+1} / (n+1)$$

$$\int \sin x dx = -\cos x$$

$$\int \cos x dx = \sin x$$

Applications:

- Area under curve
- Volume
- Work done

4. Differential Equations

Equation with derivatives.

1st Order Form:

$$dy/dx + Py = Q$$

Solution using Integrating Factor:

$$IF = e^{(\int P \, dx)}$$

Applications:

- Cooling problems
- Growth & decay
- Mechanical vibration

5. Laplace Transform (M2)

$$L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

Common Transforms:

$$L\{1\} = 1/s$$

$$L\{t\} = 1/s^2$$

$$L\{e^{at}\} = 1/(s-a)$$

Used for solving differential equations.

6. Fourier Series

Any periodic function can be written as sine & cosine series.

$$f(x) = a_0 + a_1 \cos x + b_1 \sin x + \dots$$

Used in heat transfer, vibrations.

7. Partial Derivatives

For functions with many variables.

$\partial z/\partial x$, $\partial z/\partial y$

Applications:

- Heat flow
- Fluid mechanics
- Optimization

8. Vector Calculus

Vector: Quantity with magnitude & direction.

Dot Product:

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$$

Cross Product:

$$\mathbf{A} \times \mathbf{B} = AB \sin \theta \mathbf{n}$$

Gradient, Divergence, Curl basics.

9. Complex Numbers

$$i = \sqrt{-1}$$

$$z = a + ib$$

$$\text{Magnitude} = \sqrt{a^2 + b^2}$$

Euler Form:

$$e^{i\theta} = \cos\theta + i \sin\theta$$

10. Exam Preparation Tips

- Learn formulas daily.
- Practice numericals.
- Use calculator correctly.
- Understand steps, not memorizing.
- Solve previous papers.